

What Characterizes Commonality on European NYSE Euronext Markets?

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Abstract

Using a rich dataset of orders and trades for a sample of stocks listed on the four European markets of NYSE Euronext, we apply principal component analysis and provide evidence on the existence and magnitude of commonality in returns, order flow and liquidity. We show that commonality in order flow mainly comes from foreign market members acting for their own account. Proprietary trading is a major driver in trade imbalance and return commonality. Next, we provide evidence on commonality in hidden liquidity. In contrast to commonality in visible depth that is the strongest for large firms, comovements in hidden depth seem to be stronger for small caps. We also show that commonality in returns, order flow and liquidity is not constant throughout the day. The opening of US markets is a key moment where commonality often reaches its maximum level.

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1 Introduction

Commonality refers to common cross-security variations in returns, liquidity, or order flow, and that appear within or across markets. Such comovements are important to consider, as they represent non-diversifiable sources of risk that investors have to manage. Commonality in returns is well-known since it lies at the heart of the famous Capital Asset Pricing Model. For liquidity, the emphasis from individual stocks to broader market determinants has been shifted more recently, with the seminal findings of Chordia et al. (2000), Huberman & Halka (2001), and Hasbrouck & Seppi (2001). Since then, a growing body of literature has characterized commonality in liquidity, its extension to order flow, and its relation to returns [Domowitz et al. (2005), Kumar & Lee (2006), Kaniel et al. (2008), Korajczyk & Sadka (2008)]. However, empirical evidence about what drives common cross-security variations is still limited. Prior research suggests that commonality is driven by correlated trading decisions within specific groups of investors (retail traders, institutional traders, mutual funds, etc.) but few studies attempt to assess the link between commonality and decisions within different groups of traders. And when they address this issue, the authors are often not able to directly identify trader type and rather use proxies based on trade size, firm size, or index inclusion. One exception is the recent article of Corwin & Lipson (2011), on which we partially built this paper.

Using a dataset consisting of all electronic order flow from November 1997 through February 1998 for 100 NYSE-listed stocks, Corwin & Lipson (2011) investigate the relative importance of different types of traders in driving commonality. Following the methodology of Hasbrouck & Seppi (2001), they compute the common components in four categories of order flows that are identified by trader type: program traders, non-program institutional traders, retail traders, and exchange members. The authors find that program trades and other institutional trades are the primary drivers of commonality in order flow and that these two order flow factors are significantly related to returns. They also report that commonality in professional order flow provides incremental explanatory power for returns on large firms, thereby confirming the predominant role of professional traders in large stocks' activity.

By applying the approach of Corwin & Lipson (2011) to a unique dataset on the European NYSE Euronext markets from February to March 2006, we aim at providing several contributions to the literature on commonality. Firstly, our rich dataset includes trader type identification that allows us, for each market member, not only to distinguish between *proprietary* and *client* trading but also to characterize trading as *domestic* or *foreign* based on the market member's location. These dataset features give us the opportunity to check whether comovements are primarily driven by proprietary trading, and in addition, to assess the extent to which market members' location matters in driving commonality. In other words, we address the issue of a potential

home bias in commonality. Secondly, our unique dataset allows for the construction of a very large panel of liquidity measures based on the order book, some of them including hidden depth. This gives us the opportunity to test the presence of commonality in *hidden liquidity*. Beltran-Lopez et al. (2009) show that hidden depth displays commonality within the various levels of the order book of a single stock, but to our knowledge, the existence of commonality in hidden liquidity across stocks has never been studied before. Thirdly, we investigate whether there exist intraday patterns in commonality. It is well-known that trading activity and liquidity exhibit time-of-day effects, but little is documented about the intraday evolution of commonality in returns, order flow and liquidity. To the best of our knowledge, Kempf & Mayston (2008) and Bauer (2004) are the only authors who have addressed that issue, but they only looked at commonality in liquidity. Compared to those studies, we not only analyze intraday commonality in liquidity, but also in returns and in order flow. Moreover, we make use of market members' account type and location to check whether intraday pattern of commonality differs according to those dimensions.

While several articles have shown evidence on commonality in liquidity on order-driven markets [Bauer (2004) on the Swiss Stock Exchange, Domowitz et al. (2005) on the Australian Stock Exchange, Kempf & Mayston (2008) on the Frankfurt Stock Exchange, Brockman et al. (2009) that analyze 47 exchanges around the world], we provide evidence on the existence and magnitude of commonality in returns, order flow and liquidity for a sample of stocks listed across the four national markets that form the European regulated cash market of NYSE Euronext group.¹

Consistent with prior research, we find significant commonality in returns and various trading variables. When decomposing order flow by trading type, commonality appears the strongest for order flow from foreign market members who act for their own account. In regressions aiming at assessing the importance of both account type and location in driving commonality, proprietary-foreign trading alone explains 90% of comovements in total order flow. In comparison, client-foreign trading accounts for 70.80% of total variation in order flow while client-domestic and proprietary-domestic only explain 44.8% and 22.3% of commonality respectively.

We also bring evidence of commonality for several liquidity measures based on spreads and depths in the order book. In line with previous studies, the strength of commonality in liquidity is weaker than in returns or order flow. Consistent with Corwin & Lipson (2011), we also find that the strength of commonality increases as more information from the limit order book is incorporated. Although to a lesser extent than visible liquidity, hidden liquidity seems to exhibit commonality as well, in

¹NYSE Euronext group is the holding company created in 2007 by the combination of NYSE Group, Inc. and Euronext N.V. Some years earlier, Euronext N.V. was created to bring together major marketplaces across Europe and become the first pan-European market. NYSE Euronext is today referred to as the European regulated cash market of NYSE Euronext group, with nearly 1 750 firms listed across four countries (Belgium, France, Portugal and The Netherlands).

particular for smaller stocks.

The intraday analysis reveals that commonality is clearly not constant through time and exhibits patterns, which are distinct from the patterns of the raw variables. We observe that commonality in returns especially increases after the opening of markets in the US. Commonality in trade imbalances closely follows commonality in returns, and we show that proprietary imbalances are responsible for most of this afternoon increase. Commonality in both trades and order flow displays spikes before noon and between 2:30 and 3:00 PM, which are mainly driven by proprietary foreign trading. Comovements in spreads also increase in the afternoon and reach their maximum at 3:00-3:30 PM.

The remainder of the paper is structured as follows. Section 2 provides a short review of the extant literature on commonality and its determinants. The next section describes our dataset and the sample characteristics. The methodology is presented in Section 4. We report our empirical analysis and its results in Section 5. Section 6 concludes.

2 Literature on Commonality and its Determinants

2.1 Main Empirical Evidence on Commonality

Common variations in returns across stocks have been known for a long time, since they are at the core of the Capital Asset Pricing Model [Sharpe (1964), Lintner (1965)]. The emphasis from stock-specific to systematic liquidity is far more recent, as it has been shifted in 2000. For a sample of 1169 NYSE stocks on 254 trading days, Chordia et al. (2000) estimate a market model for different liquidity measures² and obtain significant positive betas for a majority of stocks, even after controlling for individual determinants of liquidity. At the same time, Huberman & Halka (2001) analyze 254 daily observations for 240 NYSE stocks and document the presence of a systematic, time-varying component of liquidity.³

Based on the hypothesis that a set of common and idiosyncratic variables underlies both stock returns and order flows, Hasbrouck & Seppi (2001) extend the search for commonality to order flow for the 30 Dow Jones stocks. Using principal component analysis and canonical correlations, they find that both returns and order flows are characterized by common factors and that commonality in order flow explains about two-thirds of commonality in returns. Extending the approach of Chordia et al. (2000)

²They regress various firm-level liquidity proxies (the quoted and effective spreads, the proportional quoted and effective spreads, and quoted depth) on market or industry-level liquidity aggregates.

³The authors find that the innovations in the time-series liquidity proxies are positively correlated for each liquidity proxy (the quoted spread, the spread/price ratio, the quantity depth, and the dollar depth), even after controlling for individual determinants of liquidity. Their results are valid across different firm sizes and betas.

to all the stocks listed on the Stock Exchange of Hong Kong, Brockman & Chung (2002) find a significant marketwide component in both liquidity and order flow as well as an industrywide component but of smaller magnitude.

Domowitz et al. (2005) use simulations and Australian Stock Exchange data to show that order-type (i.e. market or limit orders) comovements lead to commonality in liquidity while order flow (i.e. order direction and size) comovements lead to commonality in returns.⁴ They conclude that some stocks may have little return correlation but high liquidity commonality, and vice versa. These authors also highlight the downward risk of liquidity commonality as it reveals itself more through deterioration than improvement across stocks.⁵

Using intraday data for 1986 and 1996, Harford & Kaul (2005) compare commonality in order flow, returns and trading costs for S&P and non-S&P stocks. Their findings emphasize the importance of indexing in driving commonality in order flow and returns. Kamara et al. (2008) study the evolution of liquidity commonality in a cross-section of US stocks from 1963 through 2005. Building on the market model wherein they use Amihud (2002)'s illiquidity measure,⁶ they find that commonality in liquidity has significantly increased for large firms but declined for small firms. The authors relate this result to the prevalence of both institutional investing and index trading in large stocks.

Regardless of the securities investigated, all the aforementioned papers reveal that firm-level liquidity is sensitive to liquidity aggregated at the market-level, the industry-level, or the index-level. To address commonality across exchanges, Brockman et al. (2009) analyze intraday spreads and depths on a sample of 47 exchanges across the world. They report that local sources of commonality represent about 39% of the firm's total commonality while global sources account for an additional 19%.⁷

2.2 Determinants of Commonality

The traditional theory relates commonality to comovements in underlying fundamentals. However, inconclusive empirical evidence has paved the way for friction-based theories, which tend to relate commonality to correlated decisions within specific groups of investors (retail traders, institutional traders, mutual funds, ...). According to Barberis et al. (2005), we can classify these theories into three groups, with respect to which type of market frictions they rely on.

⁴Market orders always consume liquidity, regardless of their direction and size, while limit orders always supply it. Buy (sell) orders tend to push price up (down) despite being limit or market orders.

⁵They document the asymmetry between the likelihood of liquidity movements and comovements conditional on extreme up and down markets.

⁶Their daily liquidity measure is the log of the ratio of the absolute value of daily return over the dollar volume.

⁷The authors extend the methodology of Chordia et al. (2000) to measure the impact of changes in global liquidity on changes in aggregate exchange-level liquidity.

First, style investing could drive commonality. This theory is proposed by Barberis & Schleifer (2003) to explain comovements in the returns of small stocks and value stocks as well as comovements when stocks are added to indices such as the S&P 500.⁸ The second theory is based on habitat investing (i.e., preference of many investors to trade only a subset of all available securities) and states that when investors' risk aversion, sentiment or liquidity needs change, their resulting trades may lead to common factors in the returns of the securities they hold.⁹ Thirdly, commonality could be driven by information diffusion. Because of market frictions, the prices of different securities are expected to incorporate information at different rates.¹⁰ Accordingly, comovements in returns or liquidity for stocks that incorporate information at similar rates could appear.

Style investing or information effects are expected to be predominant for professional traders (institutions, mutual funds, etc.) while habitat investing or sentiment effects are more likely to be relevant for retail investors. Empirical evidence reported in Vijh (1994), Barberis et al. (2005), Harford & Kaul (2005) or Kamara et al. (2008) gives some support to these expectations. However, most of the time evidence is indirect because the authors are not able to really identify trader type¹¹ and are constrained to use proxies based on trade size, firm size, or index inclusion. To date, only a few papers use direct trader type identification.

Kumar & Lee (2006) and Kaniel et al. (2008) focus on retail trading and provide mixed results. Using more than 1.85 million trades made by retail clients at a large US discount brokerage firm, Kumar & Lee (2006) document (i) a strong positive correlation in the buy-sell imbalance of retail investors across different portfolios of stocks¹² and (ii) a correlated trading behavior across different individuals, i.e., when one set of retail investors buys (sells) stocks, a different set of retail investors also tends to buy (sell) stocks. Their results show that retail trades do aggregate across individuals and can influence stock returns. In the opposite, Kaniel et al. (2008), who conduct a principal component analysis of net individual trading on a large cross-section of NYSE stocks, do not find strong evidence of correlated actions of individuals across stocks.

⁸The authors develop an asset pricing model in which investors allocate and move funds across styles based on relative past performance. Their model suggests that, when investors are noise traders with correlated sentiment, their correlated reallocation decisions induce comovements in returns for assets in the same style, even when their cash flows are uncorrelated.

⁹Lee et al. (1991) test the *investor sentiment hypothesis*, which assumes that movements in security prices may be attributable to changes in investor sentiment. They found that discounts on various funds move together and are correlated with prices of small stocks, suggesting that these comovements appear because both closed-end funds and small stocks tend to be held by individual investors. The authors suggest that other sentiment measures may exist (such as an institutional investor sentiment) and affect security prices.

¹⁰For example, some securities that are less costly to trade or that are held by investors with better access to breaking news may react quicker than others to new information.

¹¹Such piece of information is often not available in public market data and having access to data from brokerage firms is not always possible.

¹²When retail investors buy (sell) one group of stocks, they tend to sell (buy) other groups of stocks.

In a more recent paper, Corwin & Lipson (2011) investigate the relative importance of different types of traders (program traders, non-program institutional traders, retail traders, and exchange members) in driving commonality on a sample of 100 NYSE-listed stocks. They find that program trades and other institutional trades are the primary drivers of commonality in order flow and that these two order flow factors are significantly related to returns. Their results suggest that commonality is driven by the correlated trading decisions of professional traders, and not by correlated trading among retail investors.

We complement those studies by assessing the importance of account type – proprietary *versus* client – and location – domestic *versus* foreign – in driving commonality. The account type distinction in our dataset is highly similar to the classification that is used in Anand & Subrahmanyam (2008). Those authors use data from the Toronto Stock Exchange, a market that presents many similarities with the NYSE Euronext platform, and analyze the informational advantage of intermediaries in an electronic order-driven environment. Distinguishing between market members trading for their public customers and trading for their firm’s account, they show that the latter accounts for the majority of price discovery, though it only represents a minority of trades. However, they do not provide any result related to the contribution of varying account types to commonality.

3 Data and Sample Characteristics

The NYSE Euronext unique platform for cash trading is an electronic order-driven system where transactions result from the crossing of buy and sell orders. All retail and institutional order flow is routed through markets members, who can act either as brokers when submitting orders on behalf of clients (institutional and individual) or as dealers when trading for their own account.¹³ Three kinds of orders are most often used: (i) *limit orders*, that specify a limit price above (under) which the investor is not ready to buy (sell); (ii) *market-to-limit orders*, which do not specify any limit price, but are treated as limit orders with a price equal to the best price available at the opposite side of the market at the time they are submitted; (iii) *market orders*, which do not specify any limit price and walk up the book to exhaust liquidity until they are fully executed. Whenever an order is submitted at a price which is equal to or better than the best opposite quote, transaction takes place. Orders that are not immediately fully executed are recorded in the order book, for which the system enforces price-time

¹³For some illiquid stocks, they can also act as liquidity providers. A liquidity provider has an agreement with the Exchange authorities or with listed companies themselves in order to supply liquidity during the whole trading session; she must submit buy and sell orders in order to guarantee that the spread does not exceed a given level. However, this is not relevant for our analysis since our sample only contains liquid stocks.

priority.

Trading for most of the liquid stocks happens in various phases throughout the day. Continuous trading starts just after the opening call auction around 9:00 AM and ends at 5:25 PM.¹⁴ The market is quite transparent, as it diffuses at any time all the quotes at each side of the order book, with the associated displayed depth. Transactions (price and quantity) are also disclosed. However, two main features make the market not completely transparent. First, traders can use hidden orders for which only some fraction of the total quantity is disclosed to other market participants. Accordingly, hidden quantity becomes visible when the displayed quantity is executed and it keeps price priority but loses time priority. Second, the market is anonymous: traders do not know who places orders in the book (pre-trade anonymity), nor do they know who are both parties involved in transactions (post-trade anonymity).

Our sample contains 89 stocks of the Euronext 100 index, which is a global index including the largest and most liquid stocks listed on the NYSE Euronext market, for the period from February through April 2006 (61 trading days). All these stocks are domestic: 56 stocks are French, 16 stocks are Dutch, 11 stocks are Belgian and the remaining 6 stocks are Portuguese. They are all traded in continuous and each stock had at least one trade per day during the three-month period.

For each stock, our dataset includes common public market data about both orders and trades (price, total size, hidden size, submission/execution time-stamp, etc.) as well as additional private information. In particular, we have an identification code for the market member placing the order. We also know for which account type he is trading, which allows us to distinguish between *client trading* and *proprietary trading*. Furthermore, the country in which the market member is located is available in our database. We are then able to flag each order/trade as *domestic* or *foreign*, based on a comparison between the stock nationality and the market member's location. Table 1 gives a global picture of our trading type identification.

We only focus on the continuous trading period and our rich dataset allowed us to reprogram the market algorithm¹⁵ and get accurate order book data, including aggregate both displayed and hidden quantities available at each quote. Finally, our privileged information about orders and trades make us able to identify for each trade which order triggered it. Each trade can then be flagged according to the characteristics of the triggering order on the following three dimensions: account type (proprietary/client), location (domestic/foreign) and direction (buy/sell). For instance, if the

¹⁴A pre-opening period starts at 7:15 AM, where traders can place, modify or cancel orders without any transaction taking place. Trading itself starts at 9:00 AM with the opening call auction, where all crossing orders placed during pre-opening are executed at a single opening price. A 5-minute pre-closing period precedes the closing call auction at 5:30 PM. Finally, during the "trading at last" period, traders can trade at the price determined by the closing auction.

¹⁵Based on both public and private data, we developed a program rebuilding the limit order book second by second. Within this program, the state of the order book is updated whenever a new order is submitted or a standing limit order is modified or canceled.

triggering order has been submitted for a client, the resulting trade is considered as “client-initiated”. The same reasoning holds for the other two dimensions.

Summary statistics for the sample are provided in Table 2. Since trading activity and liquidity vary significantly by firm size, we present results for the full sample and for subsamples based on market capitalization quartiles and stock nationality respectively. Small stocks are those in the lower quartile (1) while large stocks are those in the upper quartile (4). In Panel A, we provide firm characteristics. Market capitalization is measured at the start of the sample period. All other variables are computed across all trading days in the three-month period. Market capitalization averages €19.1 billion in the full sample and ranges from €4.4 billion for small stocks to €48.2 billion for large stocks. Unsurprisingly, large caps are mainly French and Dutch stocks while small caps are rather Belgian and Portuguese. Average price is about €53 in the full sample but it drops to less than €10 in the Portuguese subsample. As expected, large stocks exhibit higher trading activity than small stocks. Daily returns are close to zero, except for the Portuguese stocks, and the standard deviation of daily returns tends to be somewhat lower for large caps.

Panel B of Table 2 documents the trading activity of market members (MM henceforth). For the full sample, we count on average 48 domestic and 94 foreign MM per stock. Both numbers increase unsurprisingly with market capitalization. Average market shares per MM are also given. We calculate market share in trades as the number of trades in which the MM is involved (as buyer or seller) divided by the total number of trades. We define market share in volumes as the aggregate volume traded by the MM (as buyer and seller) divided by the total traded volume. Both approaches lead to similar results. In the full sample, market share in trades averages 1.67% per domestic MM and 1.49% per foreign MM. These market shares are slightly smaller for larger caps that are on average traded by more MM. If we focus on the Portuguese stocks, we observe higher market shares, which is consistent with a lower trading activity and a smaller number of MM per stock. Finally, Panel B reports the proportion of proprietary trades and volumes per MM. For the full sample, the proportion of proprietary trades averages 34.63% per domestic MM and 73.01% per foreign MM. These results do not vary a lot across market capitalization quartiles and suggest that foreign MM mainly trade for their own account. The conclusion is similar when we consider the proportions of proprietary volumes.

The characteristics of order flow are presented in Panel C of Table 2. On average, 4 809 orders are submitted per day and stock in the full sample. As expected, order flow increases with market capitalization. Average order size is about 2 000 shares in the full sample. The Portuguese stocks incur however larger average order sizes, most likely due to their lower average prices. If we focus on proprietary orders, we count on average 2 106 orders per day and stock, suggesting that proprietary orders account for

slightly less than half of all order submissions. Results on average proprietary order size do not really differ from what we observe for the whole order flow. When we distinguish between domestic and foreign orders, we observe both a higher daily number of orders and a larger average order size for foreign MM. These results hold for any subsample, except the Portuguese stocks.

Panel D of Table 2 describes liquidity for the sample. For each stock, liquidity measures are averaged across all 15-minute intervals during the three-month period. In the full sample, relative spread averages 0.1% or 0.7% when computed at the 5 best quotes as $\frac{Ask_5 - Bid_5}{\frac{Ask_5 + Bid_5}{2}}$, where Ask_5 and Bid_5 are the fifth quotes in the order book. As expected, both measures depend on market capitalization. When we focus on the best quotes of the order book, displayed depth accounts on average for 41 363 shares while hidden depth averages 9 769 shares, that is to say almost 20% of the quantity available at the best prices. If we walk up the order book and consider the 5 best quotes, both displayed and hidden depths are obviously larger, with hidden liquidity that accounts on average for about 16%. We should point out that means and medians greatly differ for each depth measure, showing a substantial diversity across stocks in any subsample.

4 Methodology

We use the principal component analysis (PCA henceforth) that has been proposed by Hasbrouck & Seppi (2001), and which has been adopted by others since then (like Corwin & Lipson (2011)) to measure commonality in returns, order flow and liquidity. PCA is basically a descriptive method that relies on the variance-covariance structure of the raw variables to create new variables that should “better” represent the original dataset. By contrast to the market model, this method has the advantage that it does not require any distributional assumption on the data, and that it does not impose any structure on the common components. Its main drawback is that the common factors do not have an unambiguous economic interpretation.

Let x_{it} denote the variable (return, liquidity measure, order flow measure) observed for stock i at time t . If our sample contains N stocks and T periods, then the original dataset can be represented by a matrix $X = [X_1, X_2, \dots, X_N]$ where X_i is a vector of length T . The objective of PCA is to create new variables that we denote F_j , which are called “principal factors”. Those variables are linear combinations of the original variables, that is

$$F_j = u_{1j}X_1 + u_{2j}X_2 + \dots + u_{Nj}X_N. \quad (1)$$

The number of principal factors equals the number of original variables, N in our case.

The first principal component is built in such a way that it maximizes its explanation power, i.e., that it maximizes the variance along its axis. The second factor is orthogonal to the first factor, and maximizes the variance along its axis, and so on.

The total variance of the new dataset is exactly equal to the total variance of the raw dataset, that is: $\sum_{i=1}^N \text{Var}(X_i) = \sum_{j=1}^N \text{Var}(F_j)$.

The principal factors are obtained through the extraction of eigenvalues and eigenvectors from the variance-covariance matrix of the original dataset. λ_1 , which is the first (i.e., the highest) eigenvalue, then corresponds to the variance of the first principal factor and $u_1 = (u_{11}, u_{21}, \dots, u_{N1})$ is the associated eigenvector. λ_2 , the second highest eigenvalue, equals the variance of the second principal factor, and so on. The importance of a principal factor F_j can be then assessed through the proportion of total variance that it explains, that is by: $\frac{\lambda_j}{\sum_{j=1}^N \lambda_j}$. The higher the ratio, the more important the factor.

It is usual in PCA to standardize the original variables. Given that each of the standardized variable has a unit variance, the total variance of the dataset then equals the total number of variables, N . The proportion of variance explained by a principal factor F_j then becomes $\frac{\lambda_j}{N}$. If data were obtained randomly and the principal factors would not give new information, then they would all have variance equal to 1. So $\frac{1}{N}$ is a natural point of comparison to assess the importance of a factor: factors that explain more than $\frac{1}{N}$ might be interesting to consider.

In our analyses, we divide each trading day in 15-minute intervals. We drop all observations during the opening and closing call auctions. Given that continuous trading starts at 9:00 am and ends at 5:25 pm, we thus have 34 intervals for each day.¹⁶ This means that the length T of each vector equals 2074 (34 intervals for each of our 61 trading days).¹⁷

Consistent with Hasbrouck & Seppi (2001) and Corwin & Lipson (2011), our standardization procedure aims at removing the well-known intraday patterns found in many of the variables under scrutiny, by using for each security the interval-specific means and standard deviations.¹⁸ Changing notation slightly, we denote x_{ipd} the value of variable x observed for security i on the interval p ($p = 1, \dots, 34$) of day d ($d = 1, \dots, 61$). The firm and interval-specific means and standard deviations are noted μ_{ip} and σ_{ip} , and the standardized variables are computed as follows:

$$z_{ipd} = \frac{x_{ipd} - \mu_{ip}}{\sigma_{ip}}.$$

As mentioned earlier, one drawback of PCA is that the principal factors do not have a clear economic interpretation. To tackle that issue, we look at the “factor loadings”, i.e., the correlations between the principal factors and the original variables. Knowing

¹⁶The last interval only lasts 10 minutes.

¹⁷We only have 2013 observations for the return variable. Indeed, return is computed as the log difference between the last quote midpoint observed in each interval. Given that we exclude inter-day returns, no return can be computed for the first interval of the day.

¹⁸If we do not remove these intraday patterns, PCA shows stronger results of commonality, but which are artificially driven by the common intraday pattern observed for most stocks.

which original variables – in our case which stocks – are positively or negatively related to each factor may help find an interpretation for those factors. In particular, the existence of commonality should translate into a (first) factor to which most stocks positively correlate. That is why we provide in the forthcoming tables the number of stocks that load positively to the first component.

5 Empirical Evidence

5.1 Commonality in Returns and Trading Variables

We begin by analyzing the importance of common factors in returns, order flow and trading variables. Results are provided in Table 3, wherein a summary of the PCA is presented for the full sample in Panel A and for large and small caps in Panels B and C, respectively. We report in the first two columns of the table both the mean and standard deviation of the raw variables across all stocks and all 15-minute intervals. PCA are performed on the standardized variables. The next columns of the table report the variance explained by the first three factors (both individually and combined). Under the assumption of no commonality, each eigenvalue would equal one and the first three principal components would explain $3/N$ of the total variation. The final column reports the number of stocks that load positively to the first component.

Strong evidence of common components appears for all variables, especially in Panels A and B. For returns, the first common factor explains 12.45% of total variation for the full sample and up to 28.45% for large caps. These values are clearly bigger than their respective $1/N$ cutoffs (i.e., 1.12% for the whole sample and 4.55% for the large stocks). In addition, the factor loadings are positive for almost all the stocks in both samples, which is consistent with real commonality in returns across stocks.

The results for the other variables provide even stronger evidence of commonality. Except for imbalances, commonality tends to be stronger for variables based on orders than for variables based on trades. For the full sample, the first principal component explains up to 15.84% (12.16%) of total variation in the number of orders (trades). For trade imbalance ($\#trades$), the first component explains 9.52% of total variation, which is somewhat more than for order imbalance ($\#orders$). Commonality appears slightly weaker for imbalances measured in shares and, though not reported here, results for imbalances based on monetary volume are similar.

Comparing Panels B and C of Table 3, we observe that commonality is the strongest for large caps. In Corwin & Lipson (2011), the authors relate the weaker commonality for small caps to professional traders who tend to be more active in large securities. Although our trading type identification is different, we check in the next subsection whether MM profile could help gain insight into this result.

As explained previously, looking at the factor loadings helps find some economic

interpretation. We rank securities according to their market capitalization, and plot the individual stock loadings on the first three components for several variables. Figure 1 provides those results for return (Panel A) and trade imbalance (Panel B). Plots are similar in both cases. Securities loadings are not only (almost) all positive on the first factor, but they are also decreasing according to market capitalization. This is consistent with Corwin & Lipson (2011), with only a slight difference. They find that loadings on the first factor for the total submission imbalance are relatively equal across stocks, while we observe the same capitalization effect as for returns.¹⁹ Focusing on loadings on the second factor, we can observe the same effect as pointed out by Corwin & Lipson (2011): large (small) firms tend to be negatively (positively) correlated to that second factor. That difference between small and large firms reminds of the SMB factor in the Fama & French (1993) three factor model. This result is more evident for returns than for imbalances. Loadings on the third factor do not reveal any relationship with market capitalization.

Figure 2, Panel A provides the same plots for the number of orders. We observe the same phenomenon for the loadings on the first factor, which are almost all positive but decrease with market capitalization. Loadings on the second and third factors do not reveal any pattern.

5.2 Commonality in Order Flow by Trading Type

The previous subsection has brought strong evidence of order flow comovements in our sample of stocks. In the literature, friction-based theories suggest that such comovements may be driven by correlated trading decisions within specific groups of investors. To address this issue in our sample, we estimate principal components for order flow variables defined separately by trading type. As mentioned earlier, our purpose is to assess whether market members' proprietary trading and location affect comovements in order flow.

For each variable, Table 4 lists the proportion of total variance explained by the first eigenvalue in the PCA, along with the number of stocks that load positively to the first component (in brackets). For informative purposes, means are also given for the raw variables, pooled across firms and time. Results are presented for the full sample in Panel A and for large and small caps in Panels B and C, respectively.

From Table 4, it is clear that commonality is the strongest for order flow from foreign MM who act for their own account. In Panel A, the explained variance by the first principal component is 16.37% for the number of orders and 8.96% for the number of trades. By contrast, the corresponding figures for client trading by domestic

¹⁹Note that they use total market and limit order imbalance as their measure, while we use total trade imbalance which is mostly influenced by market orders and marketable limit orders. This may be part of the explanation in the difference.

MM are 6.27% and 4.76% respectively. Results are similar but weaker for traded volume and order volume. When we focus on proprietary trading only, we observe that commonality is always stronger for foreign MM. This result is similar when we consider only client trading, except for the traded volume.

Results for the subsamples of large and small stocks are shown in Panels B and C of Table 4. Commonality in proprietary trading by foreign MM is evident in both subsamples, but again is the strongest for large firms. For the latter, the first principal component explains 34.91% of the total variation in the number of orders and 23.77% of the total variation in the number of trades.

The above results suggest that commonality in proprietary order flow is stronger than in client order flow, and that commonality in foreign order flow is stronger than in domestic order flow. To assess the importance of these two dimensions in explaining order flow comovements, we build on Corwin & Lipson (2011) and regress the first principal component from all order flow on account-specific or location-specific first components. In particular, we first look at trade imbalance, and estimate the following regression:

$$P_t^{CombinedImbal} = \alpha + \beta_1 P_t^{PropImbal} + \beta_2 P_t^{ClientImbal} + \epsilon_t, \quad (2)$$

where $P_t^{CombinedImbal}$ is the first principal component obtained when using total trade imbalances, $P_t^{PropImbal}$ ($P_t^{ClientImbal}$) is the first principal component obtained when using only proprietary (client) imbalances.

Results are provided in Table 5, Panel A. This table displays the adjusted R^2 from regressions where the account-specific components are used individually and then together as regressors. We observe that proprietary imbalance explains 71.33% of the commonality in trade imbalances when measured in number of trades, while client imbalance only explains 28.72% of that commonality. When combined together, the R^2 increases to 82.58%. A similar picture emerges when we consider imbalances in number of shares, though in that case client imbalance individually explains 44.59% of commonality; what remains far lower than the 72.97% R^2 we observe for proprietary imbalance. We can conclude that commonality in trade imbalances is mainly driven by proprietary trading.

We proceed in a similar way for the total number of orders, for which we use the decomposition along both account type and location. Specifically, we estimate regressions of the following form:

$$P_t^{CombinedOrders} = \alpha + \beta_1 P_t^{PropForeign} + \beta_2 P_t^{ClientForeign} + \beta_3 P_t^{ClientDomestic} + \beta_4 P_t^{PropDomestic} + \epsilon_t, \quad (3)$$

where $P_t^{CombinedOrders}$ is the first principal component obtained when using total order flow, and the independent variables are the first principal components obtained from account- and location-specific order flow.

Results are provided in Panel B of Table 5. We first provide the R^2 s of regressions when the regressors are used individually. Proprietary-foreign trading alone explains 90% of commonality in total order flow when measured in number of orders. It is followed by client-foreign trading, which accounts for 70.80% of the variation. Client-domestic and proprietary-domestic trading only explain 44.8% and 22.3% of commonality respectively. We then provide results when the account- and location-specific components are added sequentially, in decreasing order of the R^2 they individually represent. Adding client-foreign order flow to proprietary-foreign order flow increases R^2 from 90.29 to 95.17%. The addition of client-domestic trading appears as a marginal improvement to 96.74%, while proprietary-domestic order flow does not increase R^2 at all. Results from order flow measured in number of shares are quite similar, though proprietary-foreign trading is slightly less important and proprietary-domestic trading now individually explains almost 44% of commonality. To sum up, commonality in order flow is essentially driven by foreign proprietary traders.

5.3 Commonality in Liquidity

Prior research provides evidence of systematic components in several measures of liquidity. In this paper, we extend previous studies to test the presence of commonality in hidden liquidity. Beltran-Lopez et al. (2009) show that hidden depth displays commonality *within* the various levels of the order book of a single stock, but, to the best of our knowledge, commonality in hidden liquidity *across* stocks has never been investigated so far.

A summary of the PCA results for liquidity is provided in Table 6, for the full sample in Panel A and for large and small caps in Panels B and C, respectively. Similarly to the table we used in Subsection 5.1, we report in the first two columns both the mean and standard deviation of the raw variables across all stocks and all 15-minute intervals. The next columns report the variance explained by the first three factors (both individually and combined). Under the assumption of no commonality, each eigenvalue would equal one and the first three principal components would explain $3/N$ of the total variation. The final column reports the number of stocks that load positively to the first component. As in all prior analyses, principal components are estimated using standardized variables. We present evidence for the two spread measures as well as the four depth measures that were defined in Section 3. Additionally, we proxy liquidity with the Cost of a Round Trip trade (CRT). At a given point in time, the CRT for a trade size q corresponds to the difference between the cost of buying q shares and the amount received from selling these q shares, divided by the value of these q shares at the contemporaneous midpoint. In particular, we compute, for two different trade sizes (1 000 and 10 000 shares), two measures: a visible CRT that is built only on displayed quantity in the order book and a real CRT that is calculated with both displayed and

hidden depth.

Table 6 brings evidence of common components in all liquidity measures but, in line with previous research, the strength of commonality in liquidity is weaker than what we observed for returns or order flow. In Panel A, it is the relative spread at the 5 best quotes that exhibits the highest first principal component, which explains up to 7.45% of total variation. If we focus on hidden depth, we observe that the variance explained by the first principal component is higher than the respective cutoff value, suggesting that commonality also exists in hidden liquidity. However, when comparing with the results for displayed depth, we conclude that the strength of commonality in hidden liquidity is weaker than when we consider displayed liquidity. This holds either at the best quotes or at the 5 best quotes of the order book.

When we compare results between spread measures, depth measures and CRT measures, we observe that commonality appears stronger for spread and weaker for depth measures. In other words, CRT measures provide stronger evidence of comovements across stocks than depth measures. This is consistent with the construction of the CRT that combines two dimensions of liquidity (tightness and depth). Furthermore, we draw the same conclusion by looking at securities loadings on the first component. While most of the stocks are positively correlated with the first factor for both spread and CRT measures, it is not the case for depth measures.

Consistent with Corwin & Lipson (2011), we also find that the strength of commonality increases as more information from the limit order book is incorporated. For example, the variance explained by the first component for relative spreads is 6.29% and the first three principal components explain 12.24% of the variation in this standardized variable. In comparison, the variance explained by the first factor for the relative spread at the 5 best quotes equals 7.45% and the first three principal components explain 14.81% of the variation in this variable. Findings based on depth or CRT are similar.

Results for large and small caps are shown in Panels B and C of Table 6. In contrast to the findings of Corwin & Lipson (2011), our results do not suggest that commonality in displayed liquidity is stronger for small firms. For spread, displayed depth and CRT measures, the first principal component always explains a larger proportion of total variance for large caps. However, the opposite holds for our two measures of hidden depth that exhibit a stronger commonality for small caps than for large firms.

In the same way as we have done for returns, trade imbalances and total orders in Section 5.1, we try to provide some economic interpretation by looking at individual securities loadings on the first factors. Figure 2, Panel B plots those loadings for the three first factors that come out of the PCA applied on relative spread, where securities have been ranked by decreasing market capitalization. We observe that loadings on the first factor slightly decrease with market capitalization, but the effect is less pronounced

than what we observed for returns or total orders. No tendency at all emerges when we look at the loadings on the second and third factors, as was the case for the total number of orders.

5.4 Intraday Evolution of Commonality

So far, we have shown the existence of commonality in returns, order flow and liquidity – the latter to a lesser extent – in our sample of Euronext 100 stocks. We have also highlighted that some trader types are mainly responsible for such comovements. Now, we propose to analyse whether commonality is constant throughout the day or whether it displays some intraday patterns. To the best of our knowledge, two papers in the literature have addressed that issue. Kempf & Mayston (2008) analyze commonality in liquidity on the 30 stocks of the DAX index of the Frankfurt Stock Exchange. They apply the market model proposed by Chordia et al. (2000) to price impact measures and assess the strength of commonality through the regression R^2 . They divide the trading day in three subperiods (until 11 AM, between 11:30 AM and 3:30 PM, and after 4 PM) and find that commonality is clearly the highest in the morning. The second paper is from Bauer (2004) who focuses on commonality in liquidity on 19 blue-chip stocks of the Swiss Stock Exchange. Using PCA and considering 9 hourly intervals, the author finds that commonality is relatively constant throughout the day for depth measures. By contrast, commonality on spread is the lowest at noon and reaches a peak around 4 PM.

To perform our intraday analysis, we apply the same type of PCA as we have used earlier. We still use the 15-minute intervals data as before, but we now group them in 30-minute intervals and perform a PCA for each of these half-hour intervals. For instance, we use variables computed between 9 and 9:15 AM and between 9:15 and 9:30 AM, and conduct a PCA for this first half-hour interval using all 61 trading days – the length T of each vector thus equals 122. Then we repeat the analysis for the second half-hour interval, using data between 9:30 and 9:45 AM and between 9:45 and 10 AM, and so on. With this approach, we get results for 17 half-hour intervals. We should point out that we still use the standardized variables when performing the PCA, so that any intraday pattern in commonality should be distinct from the intraday pattern of the raw variables. Figures 3 to 7 provide results for the full sample. Each figure is built in a similar way. The left figure displays the evolution of the raw variable along the 34 intervals of the day. The right figure shows the result of the intraday PCA analysis, i.e., it provides for each of the 17 half-hour intervals the proportion of variance explained by the first component.

Figure 3 plots the results for midquote returns. We observe no clear pattern in the intraday evolution of the return, which is to be expected if the market is relatively efficient. However, the end of the day is characterized by a succession of positive and

negative spikes, starting at 4 PM. This happens some time after the opening of markets in the US – the NYSE and Nasdaq start trading at 3:30 PM. Let us now look at the intraday evolution of commonality in returns (right figure). In Table 3, we have seen that the first factor explains 12.15% of total variance when we use all data points. As shown on Figure 3, this value fluctuates throughout the day, with a minimum of approximately 7.5% and a maximum of around 24%. Interestingly, this evolution of commonality in returns is distinct from the evolution of return itself. Commonality is quite stable around 10% until 2 PM, where it starts to climb. After a drop between 3 and 3:30 PM, just before the opening of US markets, it rises again to reach its peak between 4 and 4:30 PM. It then decreases for the last two intervals before the end of trading. It thus appears that commonality is greater in the afternoon, especially after markets have opened in the US.

Figure 4 provides the results for the total number of trades. We observe on the left figure the usual intraday pattern, where trading activity is higher in the beginning, then slowly decreases and reaches its minimum around 1 PM before climbing for the rest of the day. We can highlight the clear spike in the first interval after the opening of US markets, and the fall observed for the last interval before the closing auction. The evolution of commonality in total number of trades is markedly different from the evolution of the number of trades itself. It slowly increases in the morning, starting at a low of 5% and experiences a sudden surge up to 25% between 11:30 AM and noon. This increase is immediately followed by a sudden decrease of the same magnitude, which goes on until 1:30 PM. Commonality then rises again, with a second important increase before the opening of US markets. It then remains quite stable until 4:30 PM, where it slowly decreases until the end of the trading day. While the opening of US markets seems to play a role for commonality in trading, it is not as important as what can be observed for returns. Moreover, the moment where commonality is the strongest happens just before noon.

In Figure 5, we plot the results for relative spreads. The intraday pattern of spread is consistent with the literature: spreads start at their highest in the morning and then slowly decrease throughout the day. However, the decrease is far more important during the first hour than during the rest of the day. We also observe a tiny marginal increase during the last interval. Commonality in spread displays a different pattern. It is highly stable around 6% from the beginning of the day until 2 PM. It then starts to rise and reaches a peak at 11% just after the opening of US markets. It then slowly decreases until the end of the day, but it remains higher than 8%. Those results are not consistent with Kempf & Mayston (2008) who show that commonality is the strongest in the morning. Consistent with Bauer (2004), we find that commonality is high after 4 PM, but we do not find any decrease around noon.²⁰

²⁰We have performed the same analysis for CRT measures and found a pattern that is highly similar

Figure 6 provides the results for trade imbalances. Panel A focuses on total trade imbalance. As expected, the intraday evolution of trade imbalance is closely related to the intraday pattern of return. The similarity is even clearer when we look at the evolution of their commonality. In Panel B, we provide the results where we distinguish between imbalances for MM-initiated trades and for client-initiated trades. On the left figure, we observe that imbalances are clearly negative for client trades. Imbalances for proprietary trades are always above imbalances for client trades and, though they are negative most of the time, they are also closer to zero. Turning now to commonality, we can make the following observations. Firstly, commonality is always higher for proprietary imbalances than for client imbalances. Secondly, both series are stable, with an increase after the opening of US markets. Finally, the increase is clearly more pronounced for proprietary imbalances, and lasts longer. Hence the increase in commonality observed for imbalances – and consequently for returns – around 4:30 PM is mainly driven by trades initiated by proprietary traders.

The final intraday analysis is performed on order submission. Results are provided in Figure 7. Panel A displays results where orders are classified according to account type. Unsurprisingly, the intraday pattern on the left side is close to what have been observed for trades on Figure 4. Client orders are more numerous than proprietary orders until 2:45 PM. Proprietary orders start to outnumber client orders in the first interval following the opening of US markets. On the right side, we observe that the evolution of order flow commonality is close to what has been shown for trades. In particular, we observe two peaks for both account type series: the first for the interval 11:30 AM-noon, and the second for the interval 2:30-3:00 PM. Commonality for proprietary orders is always higher than for client orders. Panel B provides a decomposition by location. The general pattern seen on both figures is similar to the pattern in Panel A. Foreign orders always outnumber domestic orders, while commonality for foreign orders is most of the time higher than for orders placed by domestic members.

In Panel C of Figure 7, we provide the results where we decompose order flow according to both account type and location. It is clear from the left figure that proprietary foreign orders are more numerous than the other three order types, and that they are highly responsible for the surge in order flow observed after the opening of US markets. Those orders also display the highest level of commonality throughout the whole day (except for the first and the last intervals). Interestingly, client domestic orders not only display the lowest level of commonality, but this level is also quite stable throughout the day (e.g., we do not observe the spike around noon and around 3 PM as in the other series).²¹

to what we observe for spreads. The results are available upon request.

²¹We can observe that commonality for the proprietary domestic series equals zero for three intervals. The reason is strictly technical, as we cannot actually perform the PCA for those intervals for the following reason. There are some stocks for which no order of this type (proprietary and domestic) has

We also performed the intraday analysis for subsets of stocks, distinguishing across market capitalization groups and national NYSE Euronext markets. Detailed results are not reported here²² but we should point out the main facts. As expected, the results for large caps and for French stocks are highly similar to what has been shown above. Results are also relatively similar for smaller caps. However, one main difference appears for commonality in trade imbalances: while it was higher for proprietary than for client trades in the full sample, no such distinction is present for low capitalized stocks. We can also add that commonality in spreads for small caps is very flat throughout the day.

6 Concluding Remarks

Prior studies have documented common factors in returns, order flow and liquidity on many stock exchanges around the world. While the sources of commonality are subject to debate, growing evidence suggests that it mainly originates from correlated trading decisions among specific groups of investors, thereby supporting friction-based theories.

Applying principal component analysis to a detailed dataset for 89 stocks listed on the four European markets of NYSE Euronext, we analyze commonality in returns, order flow and liquidity and investigate whether account type and location matter in determining commonality.

We find clear evidence of common factors in returns and in several order flow measures. Commonality is the strongest for most liquid stocks – the variance explained by the first component for large caps may be up to 4 times the variance explained for smaller stocks. Distinguishing between client and proprietary trading, and between domestic and foreign trading, we report evidence of commonality in all order flow categories. However, commonality is stronger in foreign and proprietary order flow. Moreover, foreign trading explains much of order flow comovements while domestic trading has a marginal contribution to overall commonality. We also find that commonality in proprietary trading is a primary driver of comovements in imbalances, which lead to commonality in returns.

We also test for the presence of common factors in liquidity, including hidden depth. Consistent with previous literature, we find that commonality in liquidity is of a smaller magnitude than for returns or order flow. Commonality in spreads is stronger than commonality in displayed depth, while commonality in CRT measures (which combine

been submitted during the considered intervals for any of the 61 trading days. The standard deviation σ_{ip} of the number of orders for those intervals and for those stock thus equals zero, and we are not able to standardize the variables and to perform the PCA. That same phenomenon was present for the results provided in Table 4, but given that we used all intervals – and not only a subset of them – we could still perform the PCA, albeit with a number of lines smaller than for other variables.

²²They are available upon request.

both spread and depth dimensions) falls somewhere in between. Comovements in hidden depth are the weakest. As opposed to what has been observed on the NYSE, commonality in liquidity is more important for liquid stocks, except for hidden liquidity.

Finally, commonality in returns, order flow and liquidity is clearly not constant throughout the day. Interestingly, the intraday evolution of commonality is markedly different from the intraday pattern of the raw variables. Commonality in order flow displays an important spike around noon, at a time when order submission is not very intensive. A key moment for all variables – returns, order flow and liquidity – is the opening of markets in the US, when commonality most often reaches its highest value. In addition, we find that commonality in foreign trading is stronger than commonality in domestic trading throughout the whole day.

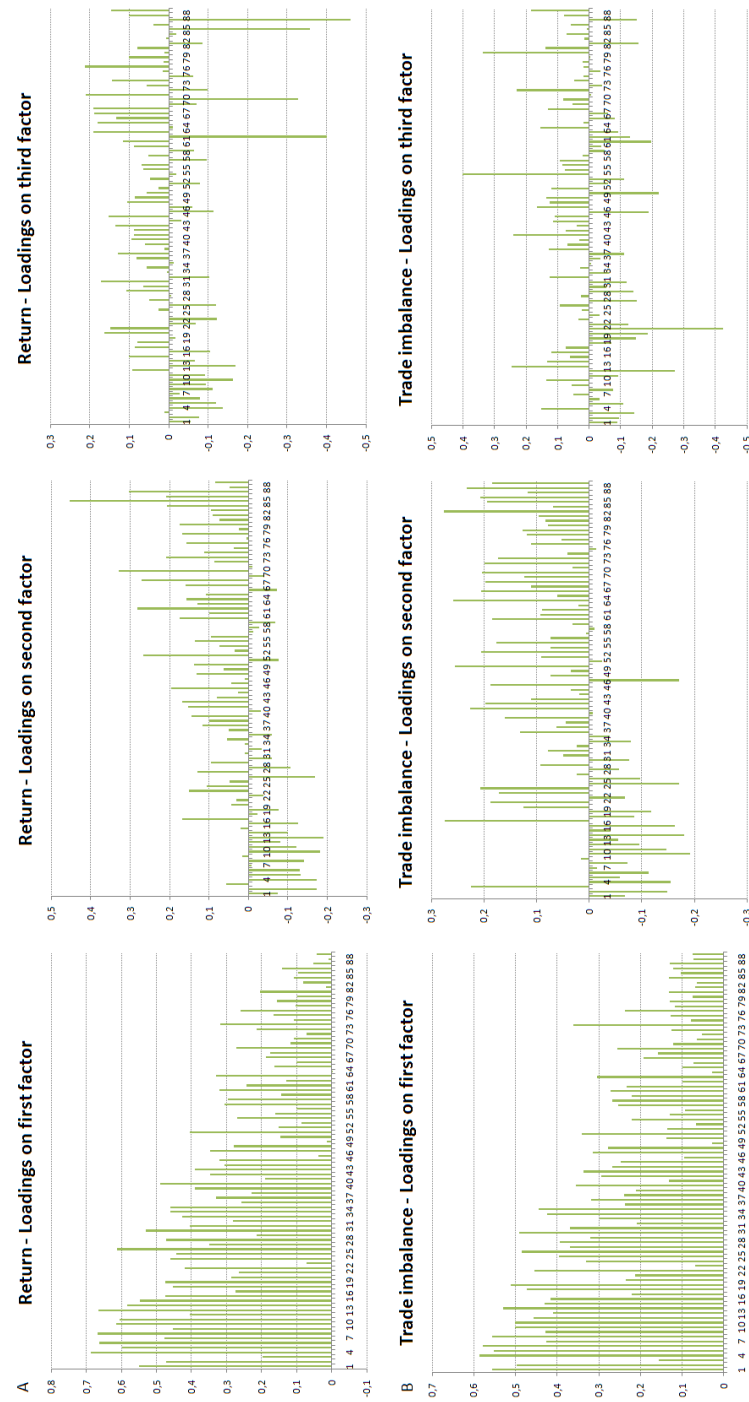
Our results complement the extant literature. Prior studies have shown that specific groups of traders are responsible for comovements across stocks. In this paper, we characterize trading according to two unexplored dimensions: account type and location. Our results suggest that this segmentation is relevant to gain insight into commonality determinants. In particular, proprietary trading from foreign market members appears to play a key role in driving commonality on NYSE Euronext.

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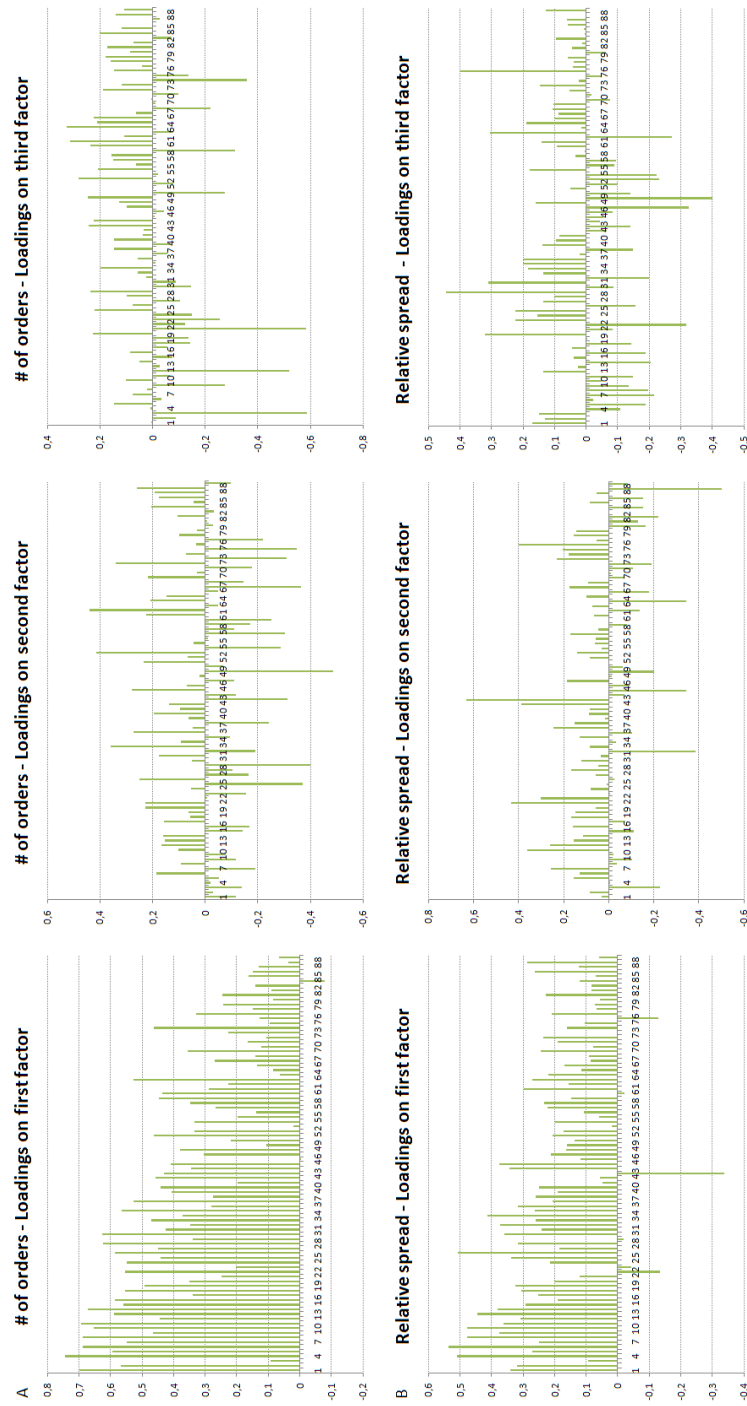
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Figure 1: Factor loadings on midquote return and trade imbalance PCA



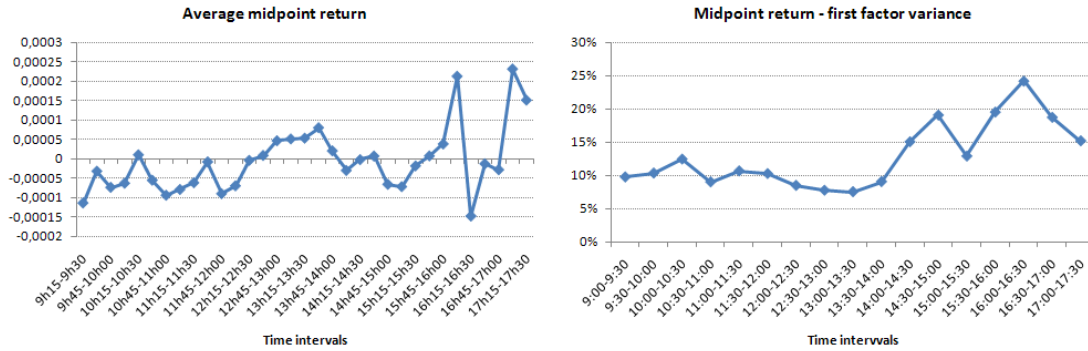
This figure shows the securities' individual factor loadings on the first three components that come out of the PCA. Securities have been ordered according to decreasing market capitalization. Panel A shows the results for PCA applied to midquote returns computed at 15-minute intervals. Panel B provides the results for PCA applied to trade imbalance $(\frac{Buy-Sell}{Buy+Sell})$ computed for every 15-minute interval.

Figure 2: Factor loadings on of orders and spread PCA



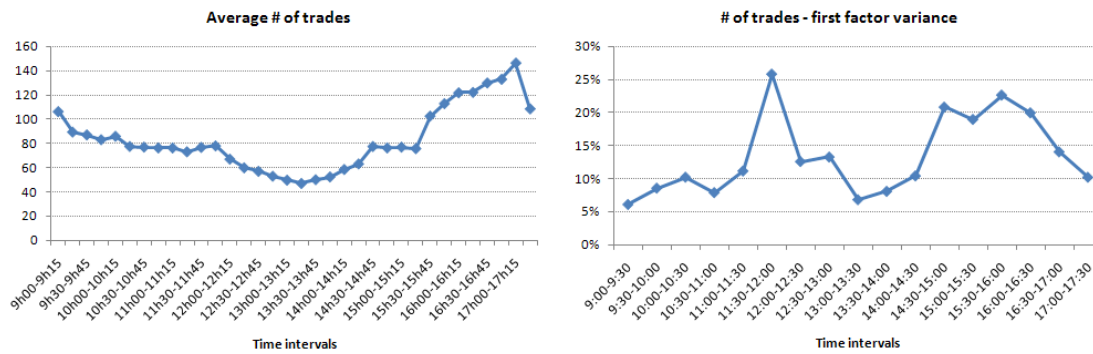
This figure shows the securities' individual factor loadings on the first three components that come out of the PCA. Securities have been ordered according to decreasing market capitalization. Panel A shows the results for PCA applied to number of orders submitted every 15-minute interval. Panel B provides the results for PCA applied to spread measured for every 15-minute interval.

Figure 3: Intraday analysis for returns



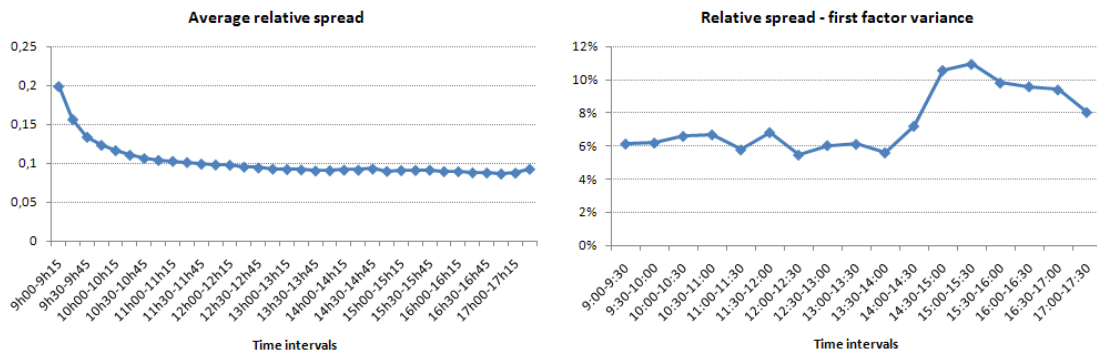
The left figure shows the intraday evolution of midquote returns for the E100 sample. For each stock and each day, we first compute returns between successive 15-minute intervals, by taking the log difference between the last midquote observed within each interval. We then take for each interval the average over all days and all stocks in our sample. The right figure displays the intraday evolution of commonality for midquote returns for the E100 sample. For each half-hour interval, we perform a PCA analysis on all returns, using all days and all stocks. The figure provides the percentage of variance taken into account by the first factor that comes out of the PCA.

Figure 4: Intraday analysis for number of trades



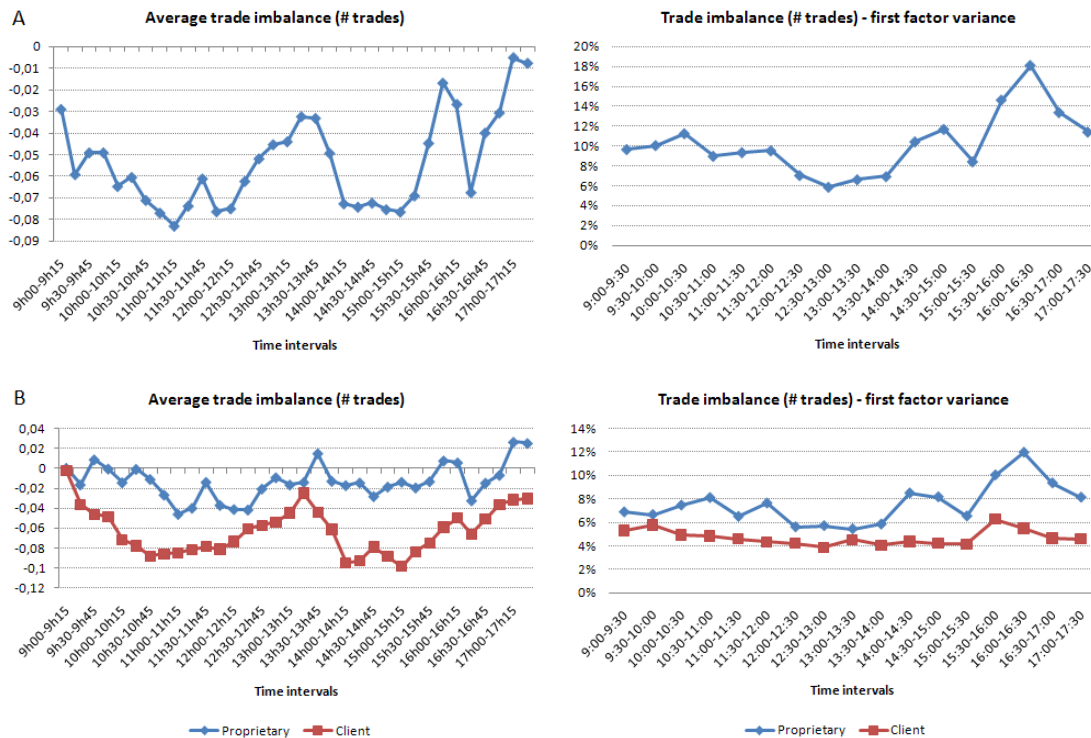
The left figure shows the intraday evolution of the number of trades for the E100 sample. For each stock and each day, we first count the number of trades observed within each 15-minute interval. We then take for each interval the average over all days and all stocks in our sample. The right figure displays the intraday evolution of commonality for the number of trades for the E100 sample. For each half-hour interval, we perform a PCA analysis on the number of trades, using all days and all stocks. The figure provides the percentage of variance taken into account by the first factor that comes out of the PCA.

Figure 5: Intraday analysis for relative spread



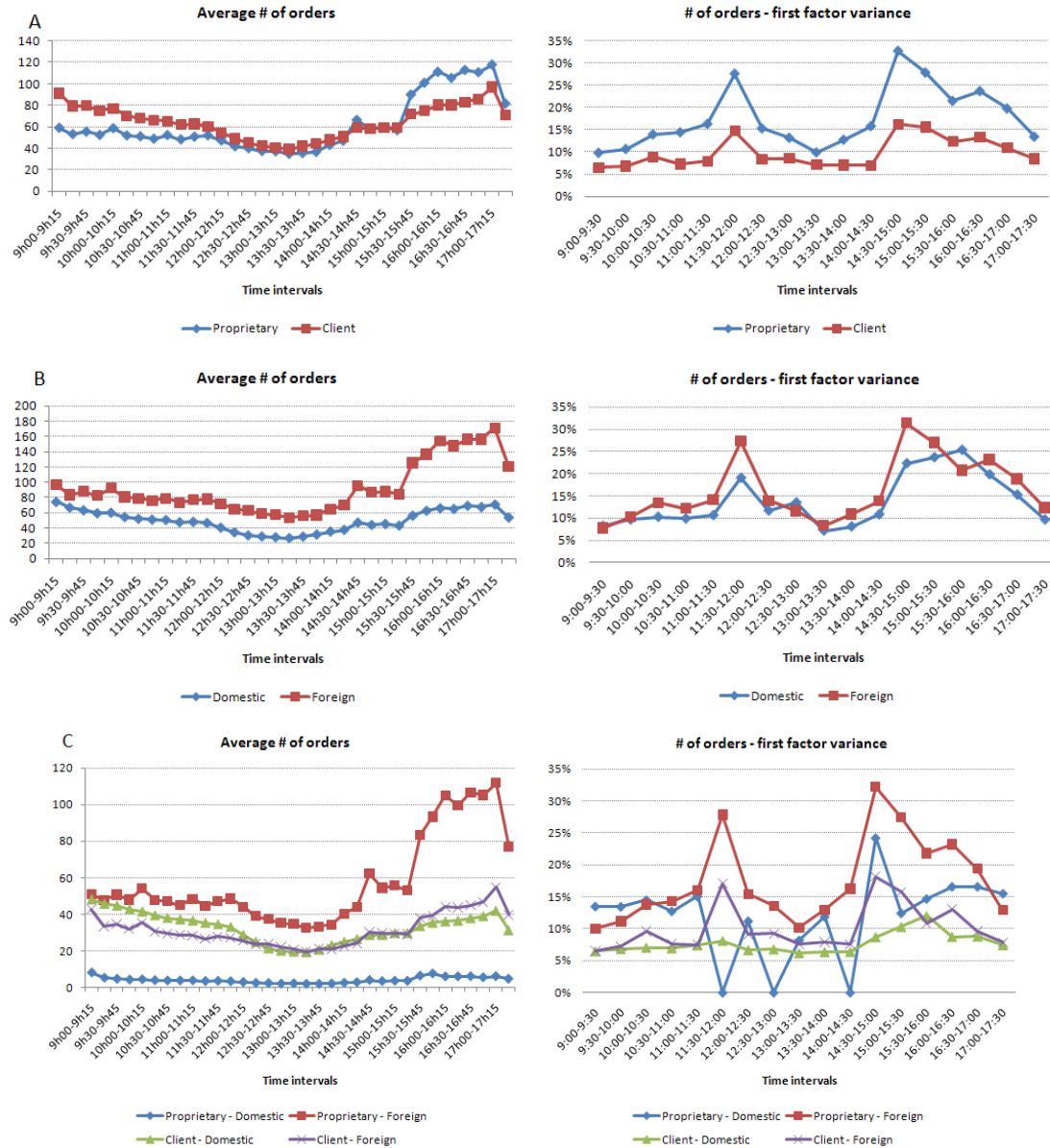
The left figure shows the intraday evolution of relative spread for the E100 sample. For each stock and each day, we first compute average spread within each 15-minute interval. We then take for each interval the average over all days and all stocks in our sample. The right figure displays the intraday evolution of commonality for relative spread for the E100 sample. For each half-hour interval, we perform a PCA analysis on relative spread, using all days and all stocks. The figure provides the percentage of variance taken into account by the first factor that comes out of the PCA.

Figure 6: Intraday analysis for trade imbalance



This figure shows the intraday analysis for trade imbalance. Panel A provides results for total trade imbalance while Panel B provides results where imbalance is decomposed according to account type (proprietary-initiated versus client-initiated). For each panel, the left figure shows the intraday evolution of trade imbalance for the E100 sample. For each stock and each day, we first measure trade imbalance within each 15-minute interval. We then take for each interval the average over all days and all stocks in our sample. The right figure displays the intraday evolution of commonality for trade imbalance for the E100 sample. For each half-hour interval, we perform a PCA analysis on trade imbalance, using all days and all stocks. The figure provides the percentage of variance taken into account by the first factor that comes out of the PCA.

Figure 7: Intraday analysis for orders by account type and location



This figure shows the intraday analysis for orders by account type and by location. Panel A provides results for orders by account type (proprietary versus client), Panel B provides results by location (domestic versus foreign) while Panel C provides results by both account type and location. For each panel, the left figure shows the intraday evolution of the number of orders for the E100 sample. For each stock and each day, we first count the number of orders within each 15-minute interval. We then take for each interval the average over all days and all stocks in our sample. The right figure displays the intraday evolution of commonality for orders for the E100 sample. For each half-hour interval, we perform a PCA analysis on orders, using all days and all stocks. The figure provides the percentage of variance taken into account by the first factor that comes out of the PCA.

Table 1: Trading type identification

Stock nationality	Proprietary trading				Client trading			
	MM's location				MM's location			
	BE	FR	NL	PT	BE	FR	NL	PT
BE	domestic	foreign	foreign	foreign	domestic	foreign	foreign	foreign
FR	foreign	domestic	foreign	foreign	foreign	domestic	foreign	foreign
NL	foreign	foreign	domestic	foreign	foreign	foreign	domestic	foreign
PT	foreign	foreign	foreign	domestic	foreign	foreign	foreign	domestic

The trading type identification that is used in this paper is summarized. *Proprietary* trading refers to orders/trades submitted/executed by market members (MM) for their own account. *Client* trading refers to orders/trades submitted/executed by MM on behalf of their clients. By comparing the stock nationality and the MM's location, we characterize trading as *domestic* or *foreign*. BE stands for Belgium, FR stands for France, NL stands for The Netherlands and PT stands for Portugal. 'Abroad' is valid for location outside these four countries.

Table 2: Sample Characteristics

	E100	1	2	3	4	BE	FR	NL	PT
Panel A: Firm characteristics									
Market capitalization (€million)	19 123 (10 232)	4 416 (4 593)	7 913 (7 750)	16 579 (15 936)	48 253 (40 924)	14 122 (7 839)	21 758 (14 117)	18 079 (9 413)	6 480 (6 323)
Average price (€)	53.41 (42.91)	46.83 (46.28)	45.29 (36.78)	63.78 (53.18)	58.05 (37.27)	59.87 (57.60)	63.25 (51.25)	31.90 (27.35)	7.10 (6.44)
Daily number of trades	2 828 (2 402)	1 362 (967)	1 864 (1 651)	2 879 (2 915)	5 272 (5 148)	1 157 (974)	3 342 (2 909)	2 928 (2 801)	831 (740)
Average trade size (#shares)	1 085 (395)	777 (352)	2 119 (372)	748 (379)	710 (601)	427 (323)	467 (358)	1 311 (1 108)	7 454 (5 199)
Daily returns (%)	-0.01 (-0.07)	0.02 (-0.05)	-0.06 (-0.08)	-0.00 (-0.06)	0.00 (-0.08)	-0.05 (-0.14)	-0.02 (-0.07)	-0.02 (-0.07)	0.19 (0.00)
Std. dev. of daily returns (%)	1.23 (1.16)	1.43 (1.29)	1.18 (1.16)	1.19 (1.20)	1.12 (1.04)	1.11 (0.93)	1.27 (1.20)	1.14 (1.16)	1.33 (1.06)
Panel B: Market Members									
Number of <i>domestic</i> MM	48 (52)	41 (45)	46 (51)	51 (54)	53 (54)	29 (29)	52 (53)	54 (56)	22 (22)
Number of <i>foreign</i> MM	94 (97)	75 (77)	83 (87)	101 (102)	118 (118)	82 (81)	97 (99)	111 (113)	41 (41)
Market share in trades per <i>domestic</i> MM (%)	1.67 (1.45)	1.99 (1.56)	1.98 (1.50)	1.33 (1.35)	1.38 (1.38)	1.64 (1.69)	1.47 (1.45)	0.93 (0.92)	5.63 (5.61)
Market share in trades per <i>foreign</i> MM (%)	1.49 (1.36)	1.88 (1.83)	1.59 (1.52)	1.38 (1.33)	1.12 (1.12)	1.91 (1.98)	1.37 (1.27)	1.44 (1.37)	2.05 (1.96)
Market share in volumes per <i>domestic</i> MM (%)	1.67 (1.37)	2.17 (1.55)	1.99 (1.52)	1.27 (1.28)	1.24 (1.19)	1.79 (1.76)	1.39 (1.35)	0.99 (0.95)	5.88 (5.83)
Market share in volumes per <i>foreign</i> MM (%)	1.49 (1.39)	1.76 (1.75)	1.58 (1.49)	1.41 (1.38)	1.19 (1.13)	1.86 (1.85)	1.40 (1.30)	1.41 (1.36)	1.84 (1.84)
% of proprietary trades per <i>domestic</i> MM	34.63 (21.15)	32.07 (19.92)	36.12 (20.82)	34.40 (21.09)	35.47 (23.35)	37.96 (36.05)	29.75 (18.18)	52.44 (51.32)	7.98 (3.26)
% of proprietary trades per <i>foreign</i> MM	73.01 (93.82)	77.50 (97.82)	74.80 (94.58)	72.84 (93.80)	68.90 (88.39)	71.70 (92.32)	75.41 (95.83)	66.98 (85.68)	68.20 (79.63)
% of proprietary volumes per <i>domestic</i> MM	35.54 (23.52)	32.50 (19.94)	36.19 (24.23)	35.40 (24.81)	37.36 (24.16)	40.67 (34.25)	30.79 (20.42)	52.19 (50.86)	10.35 (5.23)
% of proprietary volumes per <i>foreign</i> MM	71.84 (92.13)	75.94 (96.73)	73.56 (93.56)	72.00 (92.17)	67.73 (88.88)	70.14 (88.91)	74.76 (94.97)	64.74 (83.28)	64.53 (74.09)

Cross-section means (medians) are reported for each variable. The sample includes 89 stocks of the Euronext 100 index (E100). The sample period is from February through April 2006 (61 trading days). Variables are presented for the full sample and for subsamples categorized respectively on market capitalization quartiles (1 refers to small caps and 4 refers to large caps) and on stock nationality (BE stands for Belgium, FR for France, NL for The Netherlands and PT for Portugal). Panel A describes firm characteristics. Market capitalization is estimated on February 1, 2006. Panel B documents the trading activity of market members (MM). MM are flagged as *domestic (foreign)* when they are (not) located in the country of the stock. Market share in trades per MM is calculated as the number of trades in which the MM is involved (as buyer or seller) divided by the total number of trades. Market share in volumes per MM is calculated as the aggregate volume traded by the MM (as buyer and seller) divided by the total traded volume. *Proprietary* trades (volumes) are trades (volumes) executed by MM for their own account.

	E100	1	Market Capitalization			4	BE	Nationality		PT
			2	3				FR	NL	
Panel C: Order flow										
Daily number of orders	4 809 (3 957)	2 274 (1 804)	3 241 (2 803)	5 341 (4 853)	8 494 (8 176)	2 179 (1 670)	5 559 (4 654)	5 341 (4 759)		1 202 (941)
Average order size (#shares)	2 057 (853)	1 358 (735)	3 899 (804)	1 517 (767)	1 488 (1 241)	1 000 (697)	969 (730)	2 557 (2 062)		12 822 (8 179)
Daily number of <i>proprietary</i> orders	2 106 (1 717)	1 046 (904)	1 505 (1 261)	2 376 (2 237)	3 546 (3 511)	836 (721)	2 285 (1 941)	3 071 (2 659)		194 (154)
Average <i>prop.</i> order size (#shares)	2 088 (832)	1 199 (673)	4 032 (801)	1 544 (704)	1 618 (1 377)	812 (586)	1 078 (713)	2 483 (1 912)		12 808 (7 221)
Daily number of <i>domestic</i> orders	1 684 (1 137)	684 (389)	1 070 (884)	1 868 (1 538)	3 158 (2 644)	311 (270)	2 222 (1 679)	1 081 (858)		786 (607)
Average <i>domestic</i> order size (#shares)	1 942 (771)	1 375 (687)	3 672 (814)	1 468 (666)	1 280 (924)	938 (791)	777 (588)	3 276 (2 916)		11 104 (6 897)
Daily number of <i>foreign</i> orders	3 125 (2 593)	1 590 (1 372)	2 171 (1 844)	3 473 (3 216)	5 336 (5 265)	1 868 (1 381)	3 338 (2 789)	4 260 (3 885)		416 (328)
Average <i>foreign</i> order size (#shares)	2 455 (940)	1 439 (743)	5 060 (882)	1 622 (830)	1 746 (1 482)	1 009 (679)	1 167 (831)	2 384 (1 864)		17 318 (10 108)
Panel D: Liquidity										
Relative spread (%)	0.10 (0.08)	0.14 (0.12)	0.12 (0.10)	0.09 (0.08)	0.07 (0.06)	0.12 (0.09)	0.09 (0.08)	0.10 (0.09)		0.19 (0.16)
Relative spread at the 5 best quotes (%)	0.70 (0.63)	0.98 (0.91)	0.67 (0.66)	0.63 (0.60)	0.50 (0.45)	0.80 (0.74)	0.65 (0.60)	0.71 (0.65)		0.89 (0.91)
Displayed depth at the best quotes(#shares)	41 363 (4 052)	12 372 (2 830)	132 363 (3 211)	12 276 (3 689)	9 757 (7 263)	4 209 (2 684)	6 084 (3 462)	22 688 (10 159)		488 547 (48 342)
Hidden depth at the best quotes(#shares)	9 769 (1 647)	8 725 (1 156)	16 333 (1 368)	6 855 (1 651)	7 209 (2 504)	3 183 (833)	5 707 (1 661)	10 440 (2 442)		57 963 (2 400)
Displayed depth at the 5 best quotes(#shares)	219 570 (22 199)	66 144 (14 735)	683 304 (17 254)	77 351 (19 811)	58 843 (42 676)	27 277 (14 323)	34 551 (19 139)	145 084 (60 811)		2 497 576 (312 106)
Hidden depth at the 5 best quotes(#shares)	40 894 (9 775)	39 436 (8 539)	72 028 (9 018)	26 127 (9 185)	26 054 (12 818)	12 586 (6 006)	20 858 (9 094)	43 593 (14 939)		272 604 (37 657)

Cross-section means (medians) are reported for each variable. The sample includes 89 stocks of the Euronext 100 index (E100). The sample period is from February through April 2006 (61 trading days). Variables are presented for the full sample and for subsamples categorized respectively on market capitalization quartiles (1 refers to small caps and 4 refers to large caps) and on stock nationality (BE stands for Belgium, FR for France, NL for The Netherlands and PT for Portugal). Panel C reports information about order submission. *Proprietary* orders are submitted by market members for their own account. Orders are flagged as *domestic (foreign)* depending on the market member's location. Panel D reports liquidity measures. For each stock, these measures are averaged across all 15-minute intervals during the sample period. Relative spread at the 5 best quotes is defined as $\frac{Ask_5 - Bid_5}{\frac{Ask_5 + Bid_5}{2}}$, where Ask_5 and Bid_5 are the fifth price limits in the order book. Displayed (Hidden) depth at the best quotes is the sum of displayed (hidden) volume at the best bid and ask. Displayed (Hidden) depth at the 5 best quotes is the sum of aggregate displayed (hidden) volume at the 5 best limits in the order book.

Table 3: PCA results for returns and trading variables

Variable	Mean	Std. Dev.	Components			Cumulative explained variance (%)	First factor loadings
			1	2	3		
Panel A: All stocks (N=89, 1/N=1.12%)							
Midpoint return (%)	-0.00	0.21	12.15	1.85	1.65	15.65	88
Number of trades	83.17	98.96	12.16	3.7	3.46	19.31	88
Traded volume (#shares)	77 372	229 367	9.33	3.16	2.98	15.46	86
Traded volume (€)	2 195 730	4 216 652	9.2	3.18	3.01	15.39	87
Trade imbalance (#trades)	-0.05	0.4	9.52	1.81	1.73	13.06	89
Trade imbalance (#shares)	-0.02	0.47	8.48	1.7	1.57	11.75	89
Number of orders	141.43	141.42	15.84	3.78	3.31	22.93	87
Order volume (#shares)	272 887	683 565	10.92	3.08	2.74	16.74	86
Order imbalance (#orders)	-0.04	0.26	6.85	2.87	2.70	12.42	89
Order imbalance (#shares)	-0.01	0.31	4.11	1.75	1.65	7.52	87
Panel B: Large caps (N=22, 1/N=4.55%)							
Midpoint return (%)	0.00	0.20	28.45	4.95	4.59	38.00	22
Number of trades	155.07	136.38	27.23	8.00	5.14	40.36	22
Traded volume (#shares)	118 508	176 547	22.64	7.73	5.82	36.18	22
Traded volume (€)	5 013 389	6 690 531	22.36	7.68	5.91	35.95	22
Trade imbalance (#trades)	-0.09	0.31	23.09	5.50	4.68	37.27	22
Trade imbalance (#shares)	-0.04	0.37	21.09	5.18	4.64	30.91	22
Number of orders	249.81	172.78	33.95	8.09	5.00	47.05	22
Order volume (#shares)	419 517	723 504	25.64	7.23	5.14	38.00	22
Order imbalance (#orders)	-0.06	0.22	16.45	7.41	5.91	29.77	22
Order imbalance (#shares)	0.00	0.22	11.91	5.14	5.00	22.05	21
Panel C: Small caps (N=23, 1/N=4.35%)							
Midpoint return (%)	0.00	0.22	7.26	5.74	5.04	18.04	23
Number of trades	40.06	51.70	8.74	7.17	5.87	21.78	21
Traded volume (#shares)	24 634	93 504	7.91	6.61	5.83	20.35	18
Traded volume (€)	593 370	1 128 583	7.83	6.65	5.78	20.26	18
Trade imbalance (#trades)	-0.03	0.48	6.78	5.26	5.13	17.17	23
Trade imbalance (#shares)	0.00	0.56	6.26	5.13	5.13	16.52	23
Number of orders	66.89	73.36	9.13	6.96	6.35	22.43	21
Order volume (#shares)	80 958	208 710	8.22	6.52	5.91	20.65	18
Order imbalance (#orders)	-0.01	0.31	7.61	6.65	6.00	20.26	21
Order imbalance (#shares)	0.00	0.37	5.52	5.48	5.04	16.04	20

For each variable, the table lists the proportion of variance explained by the first three factors (both individually and combined) from the principal component analysis (PCA) along with the number of stocks that load positively to the first component. Under the assumption of no commonality, the variance explained by each component would equal $1/N$. All variables are calculated at 15-minute intervals from 9:00 AM through 5:25PM. In addition, all variables used in the PCA are standardized using firm and time-of-day specific means and standard deviations. Returns are defined with quote midpoints at the end of each interval. Imbalances are defined in number of trades or orders as well as in number of shares. In all cases, they are computed following the approach $\frac{Buy - Sell}{Buy + Sell}$. Means and standard deviations are for the raw variables, pooled across firms and time. The sample includes 89 stocks of the Euronext 100 index from February through April 2006. Results for the full sample are presented in Panel A. Panels B and C provide results for subsamples categorized by market capitalization, where small and large caps are defined as lower and upper quartiles, respectively.

Table 4: PCA results for order flow by trading type

Variable	Proprietary Trading		Client Trading	
	Domestic	Foreign	Domestic	Foreign
Panel A: All stocks (N=89, 1/N=1.12%)				
Number of trades	5.10% (74)	8.96% (88)	4.76% (73)	6.34% (88)
	3.52	26.23	24.81	21.99
Traded volume (#shares)	5.92% (79)	6.47% (88)	6.51% (79)	4.51% (83)
	4 835	23 649	19 979	23 963
Number of orders	12.42% (79)	16.37% (88)	6.27% (72)	8.80% (88)
	4.18	57.77	32.84	31.60
Order volume (#shares)	4.66% (79)	8.92% (86)	4.53% (80)	8.83% (85)
	16 614	95 046	57 362	79 637
Panel B: Large caps (N=22, 1/N=4.55%)				
Number of trades	9.73% (22)	23.77% (22)	12.27% (21)	13.95% (21)
	6.29	46.46	47.07	39.61
Traded volume (#shares)	9.41% (22)	16.50% (22)	15.55% (21)	11.18% (22)
	8 078	40 443	22 147	35 351
Number of orders	22.59% (22)	34.91% (22)	14.86% (20)	18.45% (21)
	7.33	96.96	61.14	54.13
Order volume (#shares)	11.09% (22)	20.00% (22)	10.55% (22)	19.59% (22)
	24 682	165 970	59 716	123 271
Panel C: Small caps (N=23, 1/N=4.35%)				
Number of trades	6.87% (17)	9.57% (23)	7.09% (20)	7.74% (23)
	1.56	13.04	13.08	11.22
Traded volume (#shares)	6.35% (19)	7.35% (22)	6.48% (19)	7.30% (18)
	1 126	6 422	9 231	7 236
Number of orders	7.65% (18)	9.17% (21)	8.70% (11)	8.96% (23)
	1.48	29.27	16.73	17.30
Order volume (#shares)	6.30% (20)	8.17% (20)	6.65% (16)	8.70% (21)
	3 556	27 324	21 554	26 110

For each variable, the table lists the proportion of total variance explained by the first eigenvalue in the principal components analysis (PCA) along with the number of stocks that load positively to the first component (in brackets). Under the assumption of no commonality, the explained variance would equal $1/N$. Principal components are estimated for order flow categorized by trader type. *Proprietary* trading refers to orders/trades submitted/initiated by market members (MM) for their own account. *Client* trading refers to orders/trades submitted/initiated by MM on behalf of their clients. Trades/Orders are flagged as *domestic* or *foreign* depending on the location of the MM who initiates/submits the trade/order. All variables are calculated at 15-minute intervals from 9:30 AM through 5:25 PM. In addition, all variables used in the PCA are standardized using firm and time-of-day specific means and standard deviations. Means are also given for the raw variables, pooled across firms and time. The sample includes 89 stocks of the Euronext 100 index from February through April 2006. Results for the full sample are presented in Panel A. Panels B and C provide results for subsamples categorized by market capitalization, where small and large caps are defined as lower and upper quartiles, respectively.

Table 5: Drivers of Trade Imbalance and Order Flow Principal Components

Panel A: Trade Imbalance				
	# of trades		# of shares	
	Individual R^2	Cumulative R^2	Individual R^2	Cumulative R^2
Proprietary	71.33%	71.33%	72.97%	72.87%
Client	28.72%	82.58%	44.59%	84.05%

Panel B: Order Flow				
	# of trades		# of shares	
	Individual R^2	Cumulative R^2	Individual R^2	Cumulative R^2
Proprietary-Foreign	90.29%	90.29%	78.70%	78.70%
Client-Foreign	70.80%	95.17%	73.82%	96.46%
Client-Domestic	44.80%	96.74%	65.98%	96.99%
Proprietary-Domestic	22.30%	96.74%	43.71%	97.08%

This table provides the adjusted R^2 for regressions of the first principal component on principal components obtained with only account- or location-specific variables. Panel A provides results for trade imbalance (measured in number of trades and in number of shares), where we distinguish between proprietary-initiated and client-initiated trades. Panel B provides results for total number of orders (measured in number of orders and in number of shares), where we distinguish according to both account type and location. In both panels, independent variables are considered either individually ("Individual R^2 " column) or are added sequentially into the regression ("Cumulative R^2 " column).

Table 6: PCA results for liquidity measures

Variable	Mean	Std. Dev.	Components			Cumulative explained variance (%)	First factor loadings
			1	2	3		
Panel A: All stocks (N=89, 1/N=1.12%)							
Relative spread (%)	0.10	0.07	6.29	3.35	2.60	12.24	83
Relative spread at the 5 best quotes (%)	0.70	0.36	7.45	4.12	3.24	14.81	76
Displayed depth at the best quotes(#shares)	41 363	262 055	4.56	4.19	3.21	11.97	46
Hidden depth at the best quotes(#shares)	9 769	77 380	2.43	2.17	2.08	6.67	45
Displayed depth at the 5 best quotes(#shares)	219 570	1 290 152	5.74	4.97	3.90	14.61	47
Hidden depth at the 5 best quotes(#shares)	40 894	259 751	3.54	2.98	2.60	9.11	39
Visible CRT for 1000 shares (%)	0.14	0.12	5.72	3.90	2.84	12.46	82
Real CRT for 1000 shares (%)	0.13	0.11	5.27	3.80	2.80	11.87	80
Visible CRT for 10 000 shares (%)	0.38	0.29	6.42	4.26	3.29	13.97	85
Real CRT for 10 000 shares (%)	0.32	0.27	5.60	3.64	3.04	12.28	86
Panel B: Large caps (N=22, 1/N=4.55%)							
Relative spread (%)	0.07	0.03	14.59	6.73	5.32	26.64	21
Relative spread at the 5 best quotes (%)	0.50	0.16	13.00	7.14	5.68	25.82	20
Displayed depth at the best quotes(#shares)	9 757	8 788	9.41	7.00	6.45	22.86	21
Hidden depth at the best quotes(#shares)	7 209	84 580	5.91	5.59	5.55	17.05	17
Displayed depth at the 5 best quotes(#shares)	58 453	49 648	9.86	8.09	7.27	25.23	21
Hidden depth at the 5 best quotes(#shares)	26 054	93 418	6.59	6.14	5.91	18.64	11
Visible CRT for 1000 shares (%)	0.08	0.04	11.73	7.18	5.32	24.23	21
Real CRT for 1000 shares (%)	0.07	0.04	11.18	7.32	5.41	23.91	22
Visible CRT for 10 000 shares (%)	0.20	0.12	13.91	8.18	6.14	28.23	21
Real CRT for 10 000 shares (%)	0.17	0.10	13.23	7.32	5.41	25.95	21
Panel C: Small caps (N=23, 1/N=4.35%)							
Relative spread (%)	0.14	0.09	7.17	6.09	5.52	18.78	14
Relative spread at the 5 best quotes (%)	0.98	0.44	9.74	8.48	6.48	24.70	15
Displayed depth at the best quotes(#shares)	13 372	69 459	7.83	6.87	5.87	20.57	15
Hidden depth at the best quotes(#shares)	8 725	75 545	6.39	5.96	5.61	17.96	14
Displayed depth at the 5 best quotes(#shares)	66 144	399 450	8.83	8.35	6.52	23.70	15
Hidden depth at the 5 best quotes(#shares)	39 436	320 840	7.22	7.13	6.52	20.87	15
Visible CRT for 1000 shares (%)	0.20	0.14	7.57	6.09	5.65	19.30	21
Real CRT for 1000 shares (%)	0.18	0.13	7.17	5.87	5.65	18.70	19
Visible CRT for 10 000 shares (%)	0.59	0.37	8.91	7.30	6.74	22.96	18
Real CRT for 10 000 shares (%)	0.50	0.35	8.17	6.83	6.04	21.04	15

For each variable, the table lists the proportion of variance explained by the first three factors (both individually and combined) from the principal component analysis (PCA) along with the number of stocks that load positively to the first component. Under the assumption of no commonality, the variance explained by each component would equal $1/N$. All variables are calculated at 15-minute intervals from 9:00 AM through 5:25 PM. In addition, all variables used in the PCA are standardized using firm and time-of-day specific means and standard deviations. Relative spread at the 5 best quotes is defined as $\frac{Ask_5 - Bid_5}{(Ask_5 + Bid_5)/2}$, where Ask_5 and Bid_5 are the fifth price limits in the order book. Displayed (Hidden) depth at the best quotes is the sum of displayed (hidden) volume at the best bid and ask. Displayed (Hidden) depth at the 5 best quotes is the sum of aggregate displayed (hidden) volume at the 5 best limits in the order book. Visible CRT is defined as the cost of a round trip trade, based on the displayed quantity available in the order book. Real CRT is defined as the cost of a round trip trade, based on the both displayed and hidden quantity available in the order book. Means and standard deviations are for the raw variables, pooled across firms and time. The sample includes 89 stocks of the Euronext 100 index from February through April 2006. Results for the full sample are presented in Panel A. Panels B and C provide results for subsamples categorized by market capitalization, where small and large caps are defined as lower and upper quartiles, respectively.